

Md Mezbahul Islam

+1 9062315296 | mailformezbah@gmail.com | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

SUMMARY

Machine Learning & Quantitative Researcher with 7+ years of experience building end-to-end models in Python (PyTorch, scikit-learn, TensorFlow) on multimodal biomedical and tabular data. PhD candidate in Computer Science (FIU) with strong background in statistics, feature engineering, and explainable AI (SHAP, GNNs, transformers). Research focuses on building interpretable models that bridge the gap between AI and clinical decision-making in precision medicine. Passionate about impactful AI, scientific research, and collaboration.

TECHNICAL SKILLS

Languages: Python (primary), C++, C, Java, R, SQL

ML / DL: PyTorch, TensorFlow, scikit-learn, Transformers, GNNs, XGBoost, SHAP, Hugging Face

Generative & Agentic AI: LLMs, RAG, PEFT/LoRA, LangChain, MCP, LLM Tool Use

Cloud & HPC: AWS (EC2, S3), Azure, HPC/SLURM, Linux, CUDA/GPU Computing

Data & MLOps: NumPy, Pandas, SciPy, JupyterLab, Git/GitHub, Pytest, GitHub Actions, FastAPI

Databases: PostgreSQL, ArangoDB, Firebase

Bioinformatics: PLINK, GWAS, multimodal clinical/genomic data analysis

PROFESSIONAL EXPERIENCE

GRADUATE RESEARCH ASSISTANT, Knight Foundation School of Computing and Information Sciences (KFSCIS), Florida International University (FIU), USA *August 2022 - Present*

- Built SHAP-based explainable ML pipelines for multimodal clinical datasets (700+ subjects), improving predictive performance by **5% over baseline** and enabling feature-level interpretability for clinicians
 - Skills:** Python, PyTorch, Scikit-learn, XGBoost, SHAP. **Accepted in IEEE BIBM 2025.**
- Developed Transformer and LLM-based explainable AI models for multimodal clinical data.
 - Skills:** Pytorch, JupyterLab, Transformers, TabExplainer, TabPFN, HuggingFace, PEFT/LoRA
- Designed and implemented a Graph Attention Network (GAT) and Graph Neural Network (GNN)
 - [MOGAT](#), IJMS 2024: Integrating multi-omics data, improving cancer subtype prediction by **5% over baseline methods**. **Skills:** PyTorch Geometric, Seaborn, Graph Data Science.
 - [ProxyExplainer](#), 41st ICML 2024: Improve subgraph-based explanations for GNNs across multiple benchmark datasets. **Skills:** CUDA, Torch, GNNExplainer.

DATA SCIENCE & INFORMATICS SCHOLAR / SUMMER INTERN, National Library of Medicine, National Institutes of Health, Bethesda, MD, USA *June 2026 – August 2026*

- Built an **agentic AI** pipeline that autonomously traverses the NLM Cell Knowledge Network to extract specific pattern for drug repurposing scope. Submitted in ICBO'2026.
 - Skills:** Agentic AI, LLM Tool Use, Iterative prompt development, Knowledge Graphs, Graph Traversal, Explainable AI.
 - Tools:** Claude + MCP, ArangoDB, Python, NetworkX, API (Open Targets, openFDA, ChEMBL).

ASSISTANT PROFESSOR, Dept. of Computer Science and Engineering (CSE), Mawlana Bhashani Science and Technology University (MBSTU), Bangladesh *April 2017 – Present (Study Leave)*

- Supervised **30+ undergraduate** and **9 MS theses** on ML/NLP/computer vision, resulting in **3 journal publications** and **2 funded projects** (UGC Bangladesh).
- Modernized curriculum by introducing Python- and ML-based coursework for 200+ CSE students annually.

LECTURER, CSE, EWU&DIU, Bangladesh

May 2015 - April 2017

SOFTWARE ENGINEER, Fronture Technologies, Bangladesh

January 2014 - April 2015

- Developed and deployed an e-commerce platform (Magento + Android) serving 1,000+ active users; optimized backend workflows and reduced checkout latency by **~30%**.
- Collaborated using Git, SQL, and REST APIs; monitored user-behavior metrics and iterated features based on data insights.

EDUCATION

PhD Candidate in Computer Science: Florida International University. *Expected December 2026*

- **Title of Dissertation:** Multi-Modal Assessment of Gene Mutation Effects on Neurocognitive Functions in Neurodegenerative Disease: Bridging Subjective and Objective Measures.
- **Concentration:** Machine learning, bioinformatics, explainable AI
- **Relevant coursework:** Advanced Topics in Machine Learning, Analysis of Algorithms.

Master of Science in Computer Science and Engineering: Jahangirnagar University, BD. *December 2014*

Bachelor of Science in Computer Science and Engineering: Jahangirnagar University, BD. *December 2013*

PUBLICATIONS

- "STEP-PD: Stage-Aware and Explainable Parkinson's Disease Severity Classification Using Multimodal Clinical Assessments", Accepted in **ICHI**, 2026.
- "REDONE-PD: Reflections of Dopamine-Related Gene Mutations on Neurocognitive Functions in Healthy Controls and Parkinson's Disease." **BIBM**, 2024.
- "Generating In-Distribution Proxy Graphs for Explaining Graph Neural Networks." In 41st **ICML**, 2024.
- Earlier ([full list](#)): SCOPE-PD'26, TILDA-X'25, MOGAT'24, IJCSIT'20; IJCCI'18; IJCSIS'18; IAJIT'17.

PROJECTS

- Created Ultrasound Password Vault leveraging ultrasonic signal encoding/decoding for secure authentication and password management; Won Blackhole Award in Winter Wonderhack 2022, MTU.
 - Store authentication information of different applications using three data fields: site URL or application name, user, and password and the master password for the application is an audio-based input which we created by embedding a password in an ultrasound audio file.
 - **Skills:** Python, tkinter, gwave and pyaudio.
- Implemented pipeline for real-time traffic-sign classification using a 50k-image (UGC-funded, 2020–21).
 - **Skills:** Python, Keras, scikit-learn, Pandas, PIL, CNNs, Kaggle dataset.
- Built a sentiment-analysis model from social-media opinion data. (UGC-funded, 2018–19).
 - **Skills:** Python, NLTK, Naive Bayes, SVM, LDA, RNN/LSTM, GPU acceleration.
- Price Prediction with Frontier & Fine-Tuned LLMs (Udemy LLM Engineering Course)
 - Built a **machine-learning pipeline** to predict product prices from short text descriptions using **fine-tuned open-source models**, demonstrating strong generalization on noisy real-world data.
 - Engineered a **text-based regression system** combining feature extraction, prompt-engineering, and supervised fine-tuning to match Frontier model performance for structured prediction.
 - **Skills:** Python, Transformers, Hugging Face, PyTorch, LangChain, RAG, Fine-tuning, FastAPI.

PRESENTATIONS

- **Posters:** FIU 4th Annual South Florida Translational Research Symposium, Port St. Lucie, FL, 2025
- **Talk:** BIBM Workshop on AI for Neurodegenerative Diseases, 2024
- **Conference:** RECOMB Computational Cancer Biology 2023

HONORS AND AWARDS

- Earned a four-year university merit scholarship (2010 - 2013)
- Placed 2nd in App Development, 5-day Boot Camp (2012)

LEADERSHIP AND EXTRACURRICULAR INVOLVEMENT

- Led the Bangladesh Student Organization (BSO) as **President** at, FIU (August 2023 - July 2024)
 - **Activity:** Organized the executive board to organize 3 flagship events with 100+ attendees each, support new students in initial settlement from airport pickup to grocery.
- Captained the BSO Soccer League 2024, FIU and achieved the **Runner-up** position
- **Assistant Proctor**, MBSTU, Bangladesh (2018 - 2019)